Homework 5 — changed numbers (extra practice)

Module 5 - Single Oral Administration · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A single 600mg dose of medication was administered orally. The medication is about 90% orally bioavailable and is characterized as having an absorption half-life of approximately 60 minutes, elimination half-life of approximately 5 hours, and apparent volume of distribution of 30 L.

1a When does the maximum concentration of drug in the plasma occur?

1b What is the expected maximum concentration following this single oral dose?

1c What is the expected AUC following this single oral dose?

1d If the dose were increased to 1200 mg, when does the maximum concentration of drug in the plasma occur?

1e If the dose were increased to 1200 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f If the dose were increased to 1200 mg, what is the AUC that you would expect?

2. A 72 kg male patient received multiple IV bolus injections of 12 mg/kg every 6 hours for 36 hours. The medication has an elimination half-life of 3 hours and an apparent

volume of distribution that is 25% of body weight.

2a What is the average concentration of drug in the plasma at steady state?

2b What is the maximum concentration of drug in the plasma at steady state?

2c What is the minimum concentration of drug in the plasma at steady state?

2d What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?